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Nonlinear Damping Compensator for Dead-Time Disturbance Voltage Considering Parameter Uncertainty in SPMSM Drives

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ABSTRACT A dynamic model of surface-mounted permanent magnet synchronous motors (SPMSMs) is constructed to include the dead-time disturbance voltage of the voltage-source inverter and SPMSM parameter uncertainty. The constructed model reveals that both the disturbance voltage and the parameter uncertainty degrade the SPMSM stator current control performance. Based on this notion, a nonlinear damping compensator is designed to simultaneously compensate for the dead-time disturbance voltage and parameter uncertainty in a unified structure. The designed compensator is advantageous because it is implemented with only the static feedback of available system variables with a low computational load, which enables the compensation signal to be calculated within one sampling period without requiring a specific calculation cycle. A rigorous analysis using a Lyapunov function shows that the designed compensator achieves the global exponential ultimate boundedness of the stator current errors with arbitrarily small bounds and fast decay rates in the presence of both the dead-time disturbance voltage and parameter uncertainty. The designed compensator performance is demonstrated by performing experiments using a 750-W laboratory SPMSM drive and comparing the experimental results with those of a conventional proportional-integral controller with a feedforward compensator.

INDEX TERMS Dead time, nonlinear damping compensator, parameter uncertainty, surface-mounted permanent magnet synchronous motor.

I. INTRODUCTION

Owing to their advantageous features, such as high precision, high power density, high efficiency, simple structure, and high torque-to-inertia ratio, surface-mounted permanent magnet synchronous motors (SPMSMs) have been widely used in many industrial applications, such as electric power steering, hybrid and electric vehicles, and ventilation and air conditioning systems. In most SPMSM drive systems, pulse-width modulation (PWM) voltage-source inverters (VSIs) are employed, and the dead time is used to prevent shoot-through over VSI switching devices. Although the dead time is relatively short compared to the PWM period, it results in various adverse effects such as fundamental voltage drops,

phase current distortions caused by low-order harmonic current, and parasitic output capacitances of semiconductor switches [1]. These non-ideal factors caused by the dead time result in VSI performance degradation and eventually lead to motor drive control performance degradation.

Two approaches have been proposed in the literature [2]–[21] to overcome the dead-time problem. In the first approach [2]–[5], a special hardware circuit is used to perform freewheeling-current polarity detection, from which the load current direction is determined, and the PWM signal is clamped to reduce the adverse dead-time effect at each switching cycle. In the second approach [6]–[21], the reference voltage command instead of the PWM signal is adjusted to compensate for the dead-time disturbance voltage. Most compensation methods employing this approach are based on the measurement of volt-second losses caused by voltage

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errors between the commanded and actual voltages in the inverter legs. The volt-second losses are averaged over each PWM carrier cycle, and the average loss values along with the phase current polarity are reflected in the voltage command to compensate for the dead-time disturbance voltage.

In [6], dead-time compensation was conducted online by using a digital filter designed to estimate the real-time current phase angle. In [7], the compensation accuracy for the dead time was improved by sampling the phase current at the dead-time interval and the beginning of the PWM period, but this requires an analog-to-digital converter (ADC) with a high conversion rate, which increases the computational complexity and hardware cost. In [8], a method based on current over-sampling was proposed, but this method is not suitable for general SPMSM drives because it uses a field-programmable gate array (FPGA) to process numerous current samples in each switching cycle.

To overcome the shortcomings of using a high conversion rate ADC [7] or a high-performance FPGA [8] for high-speed digital signal processing, in [9]–[14], feedback approaches using a current controller or disturbance observer for SPMSMs were proposed to compensate for the dead-time disturbance voltage. In [9]–[14], the dead-time disturbance voltage was compensated using only the control variable feedback signals without requiring additional hardware. Nonetheless, the methods proposed in [9]–[14] are still vulnerable to the inverter nonlinearity and system parameter uncertainty of SPMSM drives.

To address these problems, the VSI nonlinearity was studied in [15]–[21]. In [15], nonlinear characteristics were modeled to improve the VSI performance in near-zero current. In [16], a compensating signal for the dead-time disturbance voltage was constructed to cope with the VSI nonlinearity in the form of a voltage command using the rotor position and integrator output of the proportional-integral (PI) controller. In addition, other dead-time compensation methods were proposed by considering not only the dead time but also the time delay, switching signal of each phase, and current polarity in [17], by considering the nonlinear factors caused by the inverter snubber and parasitic capacitance in [18], [19], and by estimating the current ripple that strongly affects the nonlinear voltage error in [20].

On the other hand, a couple of model predictive control algorithms were proposed for the dead-time effect compensation in [21]–[23]. In [21], a dead-time compensation algorithm is proposed based on a model-based predictor that calculates and eliminates the real-time current ripple. In [22], a double vector model predictive control was proposed to compensate for the dead-time effect, in which the switching voltage vector was selected to minimize the predicted dq -axes stator current errors including the dead-time effect. In [23], the dead-time disturbance voltage was calculated based on the current polarity and switching state, and the switching voltage vector was selected to minimize the predicted dq -axes stator current errors including the calculated dead-time disturbance voltage. However, in [21]–[23],

additional model prediction control algorithms were employed to determine the compensation vector signals for the inverter nonlinearity and dead-time disturbance voltage, which requires system order increases, heavy computational load, and large memory size.

Although these studies have proposed numerous VSI nonlinearity solutions, they still cannot cope with the dead-time disturbance voltage compensation errors caused by SPMSM drive parameter uncertainty. In particular, when using the dynamic model of SPMSMs in the dead-time compensation process, parameter mismatch causes inaccurate estimation or compensation for the dead-time disturbance voltage. Regarding this problem, it was stated in [12] that performance degradation cannot be avoided without compensating for the parameter uncertainty, but it can be moderately alleviated by the current controller within its bandwidth limit.

In order to fundamentally solve this problem, both the parameter values and the dead-time disturbance voltage are estimated based on recursive least squares (RLS) adaptive filters in [24], [25] and least square method (LSM) in [26]. In [27], a model reference adaptive control (MRAC) algorithm was proposed for updating the parameter estimates in such a way as to minimize $\alpha\beta$ -axes current errors while considering VSI nonlinearity. However, these approaches suffer from the disadvantages of system order increases; complicated calculations for the RLS adaptive filters, LSM, and update laws in MRAC; and requirements of large memory size.

In this study, we consider a field-oriented control (FOC) system under the maximum torque per ampere (MTPA) strategy for SPMSMs and derive a dynamic model of SPMSMs that includes the VSI dead-time disturbance voltage and SPMSM parameter uncertainty. To simultaneously compensate for the disturbance voltage and parameter uncertainty, we propose a nonlinear damping compensator (NDC). The proposed NDC is designed in the form of a static feedback of available system variables without high-order filters, estimators, or observers, which is advantageous as it requires neither a high computational load nor a large memory size. Furthermore, the designed NDC is shown to achieve the global exponential ultimate boundedness of the current errors, and the exponential decay rates and bounds can be arbitrarily realized by adjusting the PI controller and NDC gains. The proposed NDC performance and effectiveness are demonstrated by performing extensive experiments with a 750-W laboratory SPMSM drive and comparing the experimental results with those of the typical FOC algorithm with a PI controller and feedforward compensator.

The remainder of this paper is organized as follows. In Section II, a dynamic model of SPMSMs is constructed to consider the dead-time disturbance voltage and SPMSM parameter uncertainty. In Section III, an NDC is designed for the dead-time disturbance voltage and parameter uncertainty. In Section IV, the closed-loop system stability is analyzed based on a Lyapunov function. In Section V, extensive experiments are conducted for the PI controller and the designed

NDC, and the experimental results are compared and discussed. Finally, we present our conclusions in Section VI.

II. DYNAMIC MODEL OF SPMSM

We consider an FOC system under the MTPA strategy for SPMSMs depicted in Fig. 1, where one PI controller controls the velocity in the outer loop and another PI controller along with a feedforward compensator controls the dq -axes stator currents in the inner loop, and a VSI produces SPMSM three-phase AC voltages using space vector pulse width modulation (SVPWM). The VSI dead-time disturbance voltage is shown in Fig. 1 to reflect the significant errors between the reference voltage and the actual VSI output voltage. In Fig. 1, K_e is the motor torque coefficient; J is the moment of inertia; B is the friction coefficient; ω_e^* is the reference value of electrical angular velocity, ω_e ; i_d^* and i_q^* are the reference values of dq -axes currents i_d and i_q , respectively; u_d^{pi} and u_q^{pi} are the dq -axes PI controller outputs; u_d^{ff} and u_q^{ff} are the dq -axes feedforward compensator outputs; $\tilde{u}_d$ and $\tilde{u}_q$ are the dead-time disturbance voltages in the dq -reference frame; u_d^* and u_q^* are the reference stator voltages; and u_d and u_q are the actual stator voltages.

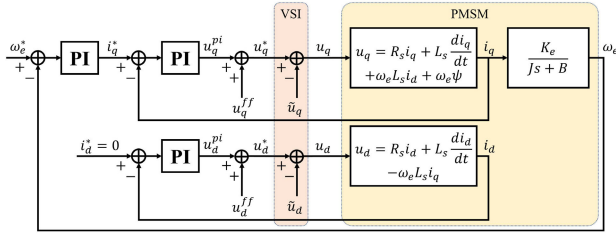


FIGURE 1. FOC system under MTPA strategy with SPMSMs, including dead-time disturbance voltage.

Ignoring cross-coupling magnetic saturation, structural asymmetry, iron losses, magnet eddy current loss, harmonics in the describing functions of the windings, rotor anisotropy, and coercive force of the magnets, the dynamic behavior of SPMSMs in the synchronously rotating dq -reference frame is described as follows [10]–[14], [21], [22], [24], [28]:

$$u_d = R_s i_d + L_s \frac{di_d}{dt} - \omega_e L_s i_q, \quad (1)$$

$$u_q = R_s i_q + L_s \frac{di_q}{dt} + \omega_e L_s i_d + \omega_e \psi, \quad (2)$$

where R_s , L_s , and ψ are the stator resistance, stator inductance, and rotor flux linkage, respectively.

Considering the dead-time disturbance voltage and parameter uncertainty, the dynamic model of SPMSMs in the synchronously rotating dq -reference frame is elaborated as follows [29]:

$$\begin{aligned} u_d^* &= \bar{R}_s i_d + \bar{L}_s \frac{di_d}{dt} - \omega_e \bar{L}_s i_q \\ &+ \Delta R_s i_d + \Delta L_s \frac{di_d}{dt} - \omega_e \Delta L_s i_q + \tilde{u}_d, \end{aligned} \quad (3)$$

$$\begin{aligned} u_q^* &= \bar{R}_s i_q + \bar{L}_s \frac{di_q}{dt} + \omega_e \bar{L}_s i_d + \omega_e \bar{\psi} \\ &+ \Delta R_s i_q + \Delta L_s \frac{di_q}{dt} + \omega_e \Delta L_s i_d + \omega_e \Delta \psi + \tilde{u}_q, \end{aligned} \quad (4)$$

where $\bar{R}_s$, $\bar{L}_s$, and $\bar{\psi}$ are the nominal values of the stator resistance, stator inductance, and rotor flux linkage, respectively; ΔR_s , ΔL_s , and $\Delta \psi$ are the parameter errors between the actual and nominal parameter values defined as $\Delta R_s = R_s - \bar{R}_s$, $\Delta L_s = L_s - \bar{L}_s$, and $\Delta \psi = \psi - \bar{\psi}$, respectively. The VSI dead-time disturbance voltage is described in the dq -reference frame as follows [16], [17], [28]:

$$\begin{bmatrix} \tilde{u}_d \\ \tilde{u}_q \end{bmatrix} = V_d \cdot \begin{bmatrix} D_d \\ D_q \end{bmatrix}, \quad (5)$$

$$\begin{bmatrix} D_d \\ D_q \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \sin \theta_e & \sin \left(\theta_e - \frac{2}{3} \pi \right) & \sin \left(\theta_e + \frac{2}{3} \pi \right) \\ \cos \theta_e & \cos \left(\theta_e - \frac{2}{3} \pi \right) & \cos \left(\theta_e + \frac{2}{3} \pi \right) \end{bmatrix} \times \begin{bmatrix} \text{sgn}(i_a) \\ \text{sgn}(i_b) \\ \text{sgn}(i_c) \end{bmatrix}, \quad (6)$$

$$V_d = \frac{T_d + T_{on} - T_{off}}{T_{pwm}} (V_{dc} - V_{sat} + V_f) + \frac{V_{sat} + V_f}{2}, \quad (7)$$

where V_d is the amplitude of the dead-time disturbance voltages, $\tilde{u}_d$ and $\tilde{u}_q$; D_d and D_q are the periodic functions determined by rotor angular position θ_e and phase current signs; T_d , T_{on} , T_{off} , and T_{pwm} are the dead time, turn-on delay time, turn-off delay time, and PWM period, respectively; V_{dc} is the dc-link voltage; V_{sat} is the switch on-state voltage drop; V_f is the diode forward voltage drop; θ_e is the electrical rotor position; i_a , i_b , and i_c are the phase currents; and sgn is the sign function. In particular, the parameters T_{on} , T_{off} , V_{sat} , and V_f are not only unmeasurable but also change depending on the variation in operating conditions; therefore, it is difficult to accurately estimate or compensate for the dead-time disturbance voltage involved with V_d .

To facilitate the following design and analysis procedure, we transform (3) and (4) such that the parameter error terms and dead-time disturbance voltages are expressed by an inner product of two vectors, that is, the inner product of the unknown part vector and the measurable variable vector:

$$\bar{L}_s \frac{di_d}{dt} = u_d^* - \bar{R}_s i_d + \omega_e \bar{L}_s i_q + \Delta_d^T \phi_d, \quad (8)$$

$$\bar{L}_s \frac{di_q}{dt} = u_q^* - \bar{R}_s i_q - \omega_e \bar{L}_s i_d - \omega_e \bar{\psi} + \Delta_q^T \phi_q, \quad (9)$$

where

$$\Delta_d = \begin{bmatrix} -\Delta R_s & -\Delta L_s & -V_d \end{bmatrix}^T, \quad (10)$$

$$\phi_d = \begin{bmatrix} i_d & \dot{i}_d - \omega_e i_q & D_d \end{bmatrix}^T, \quad (11)$$

$$\Delta_q = \begin{bmatrix} -\Delta R_s & -\Delta L_s & -V_d & -\Delta \psi \end{bmatrix}^T, \quad (12)$$

$$\phi_q = \begin{bmatrix} i_q & \dot{i}_q + \omega_e i_d & D_q & \omega_e \end{bmatrix}^T, \quad (13)$$

where the unknown parameter errors and the dead-time disturbance voltage amplitude are allowed to be time-varying but are assumed to be bounded with known positive bounds β_1 , β_2 , β_3 , and β_4 as

$$|\Delta R_s| < \beta_1, \quad |\Delta L_s| < \beta_2, \quad |V_d| < \beta_3, \quad |\Delta \psi| < \beta_4. \quad (14)$$

This assumption is reasonable because all the actual and nominal parameter values are bounded, and the dead-time disturbance voltage amplitude is bounded as shown in [15], [18]–[21], [28], [29]. In addition, it is noted that the bound estimates β_i in (14) are not necessary for the proposed NDC design, but only the existence of the bounds is enough to design and implement the NDC, which will be cleared in the following section.

III. DESIGN OF NONLINEAR DAMPING COMPENSATOR

By considering the presence of the parameter errors and dead-time disturbance voltage on the right-hand side of (3) and (4), it is found that the control performance for the dq -axes stator currents is significantly affected by both the dead-time disturbance voltage and parameter errors. Hence, it is necessary to simultaneously compensate for both the disturbance voltage and parameter errors to achieve a high degree of control performance for the dq -axes stator currents. To achieve this compensation, the conventional PI controller, along with the feedforward compensator shown in Fig. 1, can be employed. However, (3) and (4) indicate that the PI controller is insufficient for precise stator current control because the parameter error terms and disturbance voltage are time-varying, but the PI controller is only able to compensate for the DC component of time-varying errors or disturbances.

This situation requires a new compensator type, and we focus on the special structure of the parameter error terms and disturbance voltage in (3) and (4), which is in the form of the product of unknown and measurable variable parts. There are several available nonlinear compensation methods for this structure of uncertainty and disturbance as shown [30], [31], but we employ the simplest method, the NDC, in order to develop a practically useful control method that can be easily implemented with affordable hardware even at fast control cycles. The NDC can effectively and easily compensate for the uncertainty and disturbance in (3) and (4) because it is constructed using the nonlinear static feedback of available system variables without requiring system order increases, complicated computations, high-order filters, estimators, or observers [31].

Based on these critical notions related to the dynamic characteristics of SPMSM drives, we determine the reference stator voltage structure u_d^* and u_q^* with three components:

$$u_d^* = u_d^{pi} + u_d^{ff} + u_d^{nd}, \quad (15)$$

$$u_q^* = u_q^{pi} + u_q^{ff} + u_q^{nd}, \quad (16)$$

where u_d^{pi} and u_q^{pi} are the dq -axes PI controllers, u_d^{ff} and u_q^{ff} are the dq -axes feedforward compensators, and u_d^{nd} and u_q^{nd} are the dq -axes NDCs to be designed. To primarily stabilize the

stator current and compensate for the DC component of the uncertainty and disturbance, the PI controllers are designed as

$$u_d^{pi} = -c_{d1} \left(e_d + c_{d2} \int_0^t e_d d\tau \right) - \bar{L}_s c_{d2} e_d, \quad (17)$$

$$u_q^{pi} = -c_{q1} \left(e_q + c_{q2} \int_0^t e_q d\tau \right) - \bar{L}_s c_{q2} e_q, \quad (18)$$

where e_d and e_q are the stator dq -axes current errors defined as $e_d = i_d - i_d^*$ and $e_q = i_q - i_q^*$, c_{d1} , c_{d2} , c_{q1} , and c_{q2} are the positive gains, and the initial PI controller form is selected to facilitate the stability analysis in Section IV. It is notable that the proportional and integral gains can be arbitrarily set by adjusting the two independent gains c_{d1} and c_{d2} in (17) and c_{q1} and c_{q2} in (18). In other words, the actual proportional and integral gains become $-(c_{d1} + \bar{L}_s c_{d2})$ and $-c_{d1} c_{d2}$ in (17) and $-(c_{q1} + \bar{L}_s c_{q2})$ and $-c_{q1} c_{q2}$ in (18).

Similar to [32], u_d^{ff} and u_q^{ff} are designed to eliminate the stator current, coupling, and back EMF terms in (8) and (9) as

$$u_d^{ff} = \bar{R}_s i_d - \omega_e \bar{L}_s i_q, \quad (19)$$

$$u_q^{ff} = \bar{R}_s i_q + \omega_e \bar{L}_s i_d + \omega_e \bar{\psi}. \quad (20)$$

By applying the designed PI controller and feedforward compensator to (8) and (9), the closed-loop system is described in terms of the stator current errors e_d and e_q as

$$\bar{L}_s \frac{di_d}{dt} = \bar{L}_s \frac{de_d}{dt} = u_d^{nd} - c_{d1} \sigma_d - \bar{L}_s c_{d2} e_d + \Delta_d^T \phi_d, \quad (21)$$

$$\bar{L}_s \frac{di_q}{dt} = \bar{L}_s \frac{de_q}{dt} = u_q^{nd} - c_{q1} \sigma_q - \bar{L}_s c_{q2} e_q + \Delta_q^T \phi_q, \quad (22)$$

where σ_d and σ_q are introduced to simplify the notations as

$$\sigma_d = e_d + c_{d2} \int_0^t e_d d\tau, \quad (23)$$

$$\sigma_q = e_q + c_{q2} \int_0^t e_q d\tau. \quad (24)$$

Now, by employing the technique described in [31], we design the NDC as follows:

$$u_d^{nd} = - \sum_{i=1}^3 \kappa_{di} \phi_{di}^2 \sigma_d, \quad (25)$$

$$u_q^{nd} = - \sum_{i=1}^4 \kappa_{qi} \phi_{qi}^2 \sigma_q, \quad (26)$$

where κ_{di} and κ_{qi} are the positive nonlinear damping gains, and ϕ_{di} and ϕ_{qi} are the i -th elements of ϕ_d and ϕ_q defined in (11) and (13), respectively. By applying the designed NDC to (21) and (22), the final closed-loop system is described in terms of e_d and e_q as

$$\bar{L}_s \frac{de_d}{dt} = -c_{d1} \sigma_d - \bar{L}_s c_{d2} e_d + \sum_{i=1}^3 \Delta_{di} \phi_{di} - \sum_{i=1}^3 \kappa_{di} \phi_{di}^2 \sigma_d, \quad (27)$$

$$\bar{L}_s \frac{de_q}{dt} = -c_{q1}\sigma_q - \bar{L}_s c_{q2}e_q + \sum_{i=1}^4 \Delta_{qi}\phi_{qi} - \sum_{i=1}^4 \kappa_{qi}\phi_{qi}^2\sigma_q, \quad (28)$$

where Δ_{di} and Δ_{qi} are the i -th elements of Δ_d and Δ_q defined in (10) and (12), respectively.

To implement (25) and (26), we need ϕ_d and ϕ_q consisting of i_d , i_q , $\frac{di_d}{dt}$, $\frac{di_q}{dt}$, ω_e , i_d , i_b , i_c , and θ_e , which can either be measured or calculated. Hence, the NDC (25) and (26) is feasible simply by the static feedback of the available system variables without requiring any high-order filters, estimators, or observers. Compared to other compensation methods based on model predictive control [21]–[23], multiple RLC adaptive filters [24], [25], LSM [26], MRAC [27], the proposed NDC show significant practical advantages in that it can be implemented without requiring system order increases, heavy computational load, or large memory size, and the compensation signal can be calculated within one sampling period without requiring longer multiple calculation cycles. The NDC input-output structure is illustrated in Fig. 2.

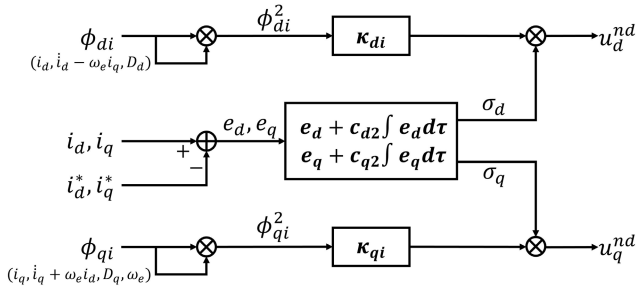


FIGURE 2. Structure of nonlinear damping compensator.

On the other hand, as shown in (25) and (26), the NDC has the quadratic terms of system variables, which are nonlinear and may lead to a side effect from large values of system variables such that the NDC output may excessively increase and cause the saturation of reference stator voltages. Hence, in order to avoid the side effect, the nonlinear damping gains κ_{di} and κ_{qi} must be tuned such that they are chosen with small values at first and then are gradually increased to find optimal gain values.

IV. STABILITY ANALYSIS

In this section, we rigorously analyze the closed-loop system stability after applying the PI controller, feedforward compensator, and NDC designed in Section III. Because the closed-loop system is nonlinear and time-varying as shown in (27) and (28), the transfer function approach cannot be used for the closed-loop stability analysis, but we employ the Lyapunov function-based approach that is well-known to be suitable for analyzing nonlinear time-varying systems. First, we derive the first-order derivatives of $\bar{L}_s\sigma_d$ and $\bar{L}_s\sigma_q$, which are used in the following mathematical development.

Using (23), (24), (27), and (28), we obtain

$$\begin{aligned} \bar{L}_s \frac{d\sigma_d}{dt} &= \bar{L}_s \frac{de_d}{dt} + \bar{L}_s c_{d2}e_d \\ &= -c_{d1}\sigma_d + \sum_{i=1}^3 \Delta_{di}\phi_{di} - \sum_{i=1}^3 \kappa_{di}\phi_{di}^2\sigma_d, \end{aligned} \quad (29)$$

$$\begin{aligned} \bar{L}_s \frac{d\sigma_q}{dt} &= \bar{L}_s \frac{de_q}{dt} + \bar{L}_s c_{q2}e_q \\ &= -c_{q1}\sigma_q + \sum_{i=1}^4 \Delta_{qi}\phi_{qi} - \sum_{i=1}^4 \kappa_{qi}\phi_{qi}^2\sigma_q. \end{aligned} \quad (30)$$

Next, a Lyapunov function is chosen as

$$V = \frac{1}{2}\bar{L}_s\sigma_d^2 + \frac{1}{2}\bar{L}_s\sigma_q^2, \quad (31)$$

and the derivative of V along the closed-loop system is derived by using (29) and (30) as

$$\begin{aligned} \frac{dV}{dt} &= \bar{L}_s\sigma_d \frac{d\sigma_d}{dt} + \bar{L}_s\sigma_q \frac{d\sigma_q}{dt} \\ &= \sigma_d \left(-c_{d1}\sigma_d + \sum_{i=1}^3 \Delta_{di}\phi_{di} - \sum_{i=1}^3 \kappa_{di}\phi_{di}^2\sigma_d \right) \\ &\quad + \sigma_q \left(-c_{q1}\sigma_q + \sum_{i=1}^4 \Delta_{qi}\phi_{qi} - \sum_{i=1}^4 \kappa_{qi}\phi_{qi}^2\sigma_q \right) \\ &= -c_{d1}\sigma_d^2 - c_{q1}\sigma_q^2 - \sum_{i=1}^3 \kappa_{di} \left(\sigma_d\phi_{di} - \frac{\Delta_{di}}{2\kappa_{di}} \right)^2 \\ &\quad - \sum_{i=1}^4 \kappa_{qi} \left(\sigma_q\phi_{qi} - \frac{\Delta_{qi}}{2\kappa_{qi}} \right)^2 + \sum_{i=1}^3 \frac{\Delta_{di}^2}{4\kappa_{di}} + \sum_{i=1}^4 \frac{\Delta_{qi}^2}{4\kappa_{qi}} \\ &\leq -c_{d1}\sigma_d^2 - c_{q1}\sigma_q^2 + \beta_\kappa, \end{aligned} \quad (32)$$

where β_κ is defined as

$$\beta_\kappa = \sum_{i=1}^3 \frac{\beta_i^2}{4\kappa_{di}} + \sum_{i=1}^4 \frac{\beta_i^2}{4\kappa_{qi}}, \quad (33)$$

where β_i is defined as the element bounds of Δ_d and Δ_q in (14). Then, because $\dot{V} < 0$ as long as $c_{d1}\sigma_d^2 + c_{q1}\sigma_q^2 > \beta_\kappa$, it is guaranteed that σ_d and σ_q globally exponentially converge to the following compact residual set:

$$S = \left\{ (\sigma_d, \sigma_q) \in \mathcal{R}^2 : c_{d1}\sigma_d^2 + c_{q1}\sigma_q^2 \leq \beta_\kappa \right\}. \quad (34)$$

Furthermore, because $\frac{dV}{dt} < 0$ as long as $|\sigma_d| > \sqrt{\frac{\beta_\kappa}{c_{d1}}}$ or $|\sigma_q| > \sqrt{\frac{\beta_\kappa}{c_{q1}}}$, it is implied that σ_d and σ_q globally exponentially converge to the following compact residual set:

$$S = \left\{ (\sigma_d, \sigma_q) \in \mathcal{R}^2 : |\sigma_d| \leq \sqrt{\frac{\beta_\kappa}{c_{d1}}} \text{ and } |\sigma_q| \leq \sqrt{\frac{\beta_\kappa}{c_{q1}}} \right\}, \quad (35)$$

from which it is guaranteed that σ_d and σ_q are exponentially ultimately bounded with arbitrarily small bounds by adjusting the PI control gains c_{di} and c_{qi} and the nonlinear damping

gains κ_{di} and κ_{qi} . In addition, (32) shows that the exponential decay rates of σ_d and σ_q are determined by the PI controller gains c_{d1} and c_{q1} ; therefore, σ_d and σ_q are globally exponentially ultimately bounded with arbitrarily fast decay rates by adjusting the PI controller gains c_{d1} and c_{q1} .

Now, we show the global exponential ultimate boundedness of e_d and e_q with arbitrarily small bounds and fast decay rates by proving the global exponential ultimate boundedness of $\int e_d d\tau$ and $\int e_q d\tau$ with arbitrarily small bounds and fast decay rates. Taking the input and output with σ_d and $\int e_d d\tau$ in (23) and σ_q and $\int e_q d\tau$ in (24) and applying the Laplace transform to (23) and (24) yield

$$\mathcal{L}\left[\int e_d d\tau\right] = \frac{1}{s + c_{d2}} \mathcal{L}[\sigma_d], \quad (36)$$

$$\mathcal{L}\left[\int e_q d\tau\right] = \frac{1}{s + c_{q2}} \mathcal{L}[\sigma_q], \quad (37)$$

where $\mathcal{L}$ denotes the Laplace transform operator. Then, (36) and (37) imply that $\int e_d d\tau$ and $\int e_q d\tau$ are globally exponentially ultimately bounded with arbitrarily small bounds and fast decay rates because c_{d2} and c_{q2} are positive, and σ_d and σ_q are already globally exponentially ultimately bounded with arbitrarily small bounds and fast decay rates. Hence, in view of (23) and (24), it is evident that e_d and e_q are also globally exponentially ultimately bounded with arbitrarily small bounds and fast decay rates by adjusting the PI controller gains c_{di} and c_{qi} and the nonlinear damping gains κ_{di} and κ_{qi} in the presence of the parameter uncertainty and dead-time disturbance voltage. Furthermore, it is noted that this remarkable result is still achieved, although the unknown parameter errors and dead-time disturbance voltage amplitude ΔR_s , ΔL_s , $\Delta \psi$, and V_d in (10) and (12) are time-varying as long as the boundedness is satisfied as in (14).

V. EXPERIMENTS

A. SETUP

We demonstrate the validity and performance of the proposed NDC by performing extensive experiments using a 750-W SPMSM drive system. The experimental setup is built with an SPMSM drive system and a load motor drive system, as shown in Fig. 3. The TMS320F28069 digital signal processor (DSP) is employed for the SPMSM drive system because it is widely used in industrial control applications owing to its high-speed real-time signal processing capability at a clock frequency of 90 MHz. The SPMSM drive system consists of an AC source, a full-bridge rectifier, a three-phase PWM inverter stage, current sensors, and an incremental encoder with a resolution of 2048 p/rev to measure the rotor position. For the VSI, an intelligent power module, MITSUBISHI PS21765, is employed, which integrates an AC 100 V-200 V three-phase inverter drive with a 600 V/20 A low-loss IGBT inverter bridge with N-side three-phase output DC-to-AC power conversion. The actual SPMSM parameter values provided by the manufacturer are listed in Table 1.

The operational period of the current control loop is set to half of the PWM period such that the current control

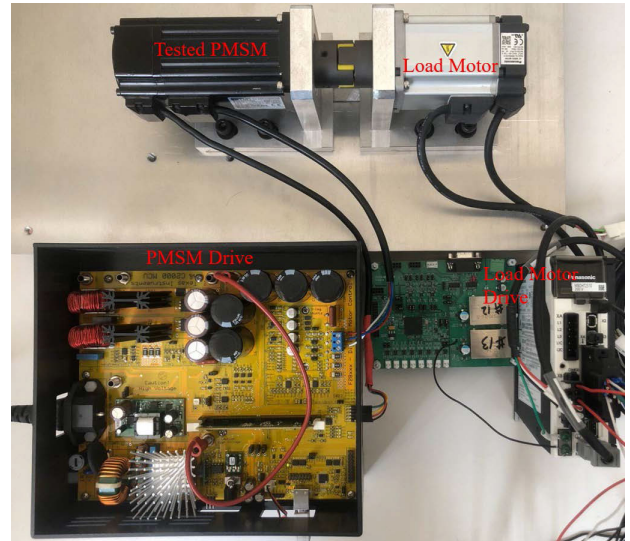


FIGURE 3. Experimental setup.

TABLE 1. SPMSM parameters.

Parameter	Value	Unit
Rated power	750	W
Rated voltage	220	V
Rated velocity	3000	r/min
Stator resistance R_s	1.02	Ω
Stator inductance L_s	8.25	mH
Rotor flux linkage ψ	0.113	Wb
Number of pole pairs p_p	4	

operation is executed twice for each PWM period. The dead time of the DSP PWM module for power switching gates is set to 3 μ s by considering the delays in the inverter drive circuits. The other drive system specifications are listed in Table 2.

TABLE 2. PWM VSI parameters.

Parameter	Value	Unit
DC link	310	V
PWM period	66	μ s
Dead time	3.0	μ s
Turn-on/off delay	180/320	ns
Saturation voltage	1.09	V
Diode forward voltage	1.3	V
IGBT/Diode R_{on}	85/32	m Ω

To achieve an acceptable current control performance, the following tuning procedure can be used for the PI current controller and the proposed NDC gains. First, the PI controller gains c_{d1} , c_{d2} , c_{q1} , and c_{q2} in (17) and (18) are tuned based on the tuning rule of [32]–[36] independent of the designed NDC. Second, all the nonlinear damping gains κ_{di} and κ_{qi} in (25) and (26) are set to small values (e.g., $\kappa_{di} = 0.001$, $\kappa_{qi} = 0.001$), and the operation is started.

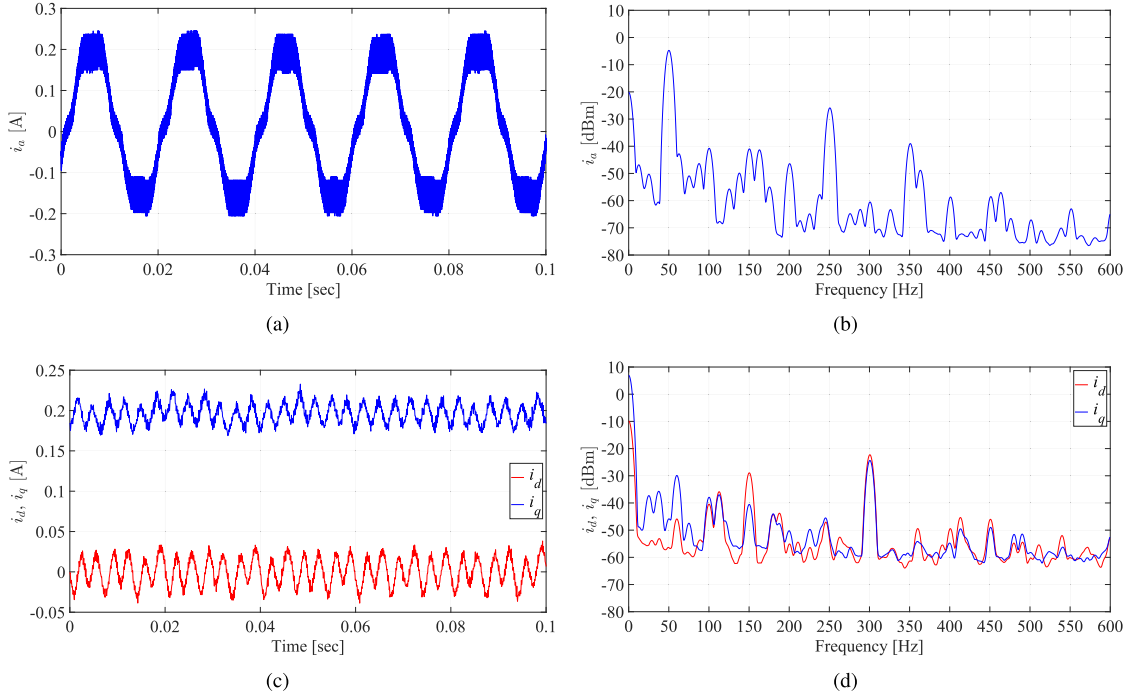


FIGURE 4. Experimental results of the first case with only PI controller and feedforward compensator: (a) a-phase current, (b) magnitude spectrum of a-phase current, (c) dq -axes currents, and (d) magnitude spectra of dq -axes currents.

Then, if the resulting performance is satisfactory with a small steady-state error (SSE) and low total harmonic distortion (THD), the tuning process is completed; otherwise, the non-linear damping gains are slightly increased. This process is repeated until satisfactory performance is achieved.

Three experimental cases are used to demonstrate the effectiveness and performance of the proposed NDC in comparison with a conventional PI controller with a feedforward compensator. In the first case, only the PI controller with the feedforward compensator is applied such that the reference stator voltages are implemented as $u_d^* = u_d^{pi} + u_d^{ff}$ and $u_q^* = u_q^{pi} + u_q^{ff}$ in (15) and (16). In the second case, the NDC only for the dead-time disturbance voltage is added to the first case such that the reference stator voltages are implemented as $u_d^* = u_d^{pi} + u_d^{ff} + u_d^{nd}$ and $u_q^* = u_q^{pi} + u_q^{ff} + u_q^{nd}$ in (15) and (16), where $u_d^{nd} = -\kappa_{d3}\phi_{d3}^2\sigma_d$ and $u_q^{nd} = -\kappa_{q3}\phi_{q3}^2\sigma_q$. In the third case, the complete NDC for both the dead-time disturbance voltage and parameter errors is added to the first case such that the reference stator voltages are implemented as $u_d^* = u_d^{pi} + u_d^{ff} + u_d^{nd}$ and $u_q^* = u_q^{pi} + u_q^{ff} + u_q^{nd}$ in (15) and (16) with (25) and (26). In all three cases, the PI controller gains are set as $c_{d1} = 0.4$, $c_{q1} = 0.5$, $c_{d2} = 10$, and $c_{q2} = 10$, the nonlinear damping gains are set as $\kappa_{d1} = \kappa_{d2} = \kappa_{d3} = 0.01$, $\kappa_{q1} = \kappa_{q2} = \kappa_{q3} = 0.01$, and $\kappa_{q4} = 0.001$, and the reference rotor speed is set to 750 r/min (i.e., 50 Hz in electrical frequency).

B. RESULTS

Figs. 4, 5, and 6 show the experimental results of the three cases. In the experiments, all signals were measured using

an RTE1034 oscilloscope from Rohde and Schwarz, which is equipped with an integrated power measurement and analysis software that enables THD measurements over time. The phase current is directly measured using a current probe, RT-ZC15B, with 50 MHz bandwidth and 30 A (RMS) maximum current, and the dq -axes currents are measured by outputting the dq -axes current values in the form of analog signals and probing the analog signals with the oscilloscope. In each of Figs. 4, 5, and 6, subfigures (a), (b), (c), and (d) show the a-phase current, magnitude spectrum of the a-phase current, dq -axes currents i_d and i_q in the synchronous reference frame, and magnitude spectra of the dq -axes currents, respectively.

In Figs. 4(a), 5(a), and 6(a), the current ripples are commonly observed, which are due to the fifth and seventh harmonic components of the rotor electrical frequency (50 Hz) in the phase current caused by the VSI dead time. These harmonic components are clearly demonstrated by the magnitude spectrum of the a-phase stator current in Figs. 4(b), 5(b), and 6(b). Furthermore, as shown in Figs. 4(c), 5(c), and 6(c), the current ripples in the phase current directly affect the control variables in the synchronous reference frame, which results in the current ripples containing the sixth and twelfth harmonic components of the 50 Hz rotor electrical frequency. These harmonic components are also demonstrated by the magnitude spectra of the dq -axes stator currents in Figs. 4(d), 5(d), and 6(d).

Comparing Figs. 4, 5, and 6, it is clear that the best and worst performances are obtained in the third and first experiment, respectively, which verifies that the designed NDC effectively compensates for both the dead-time disturbance

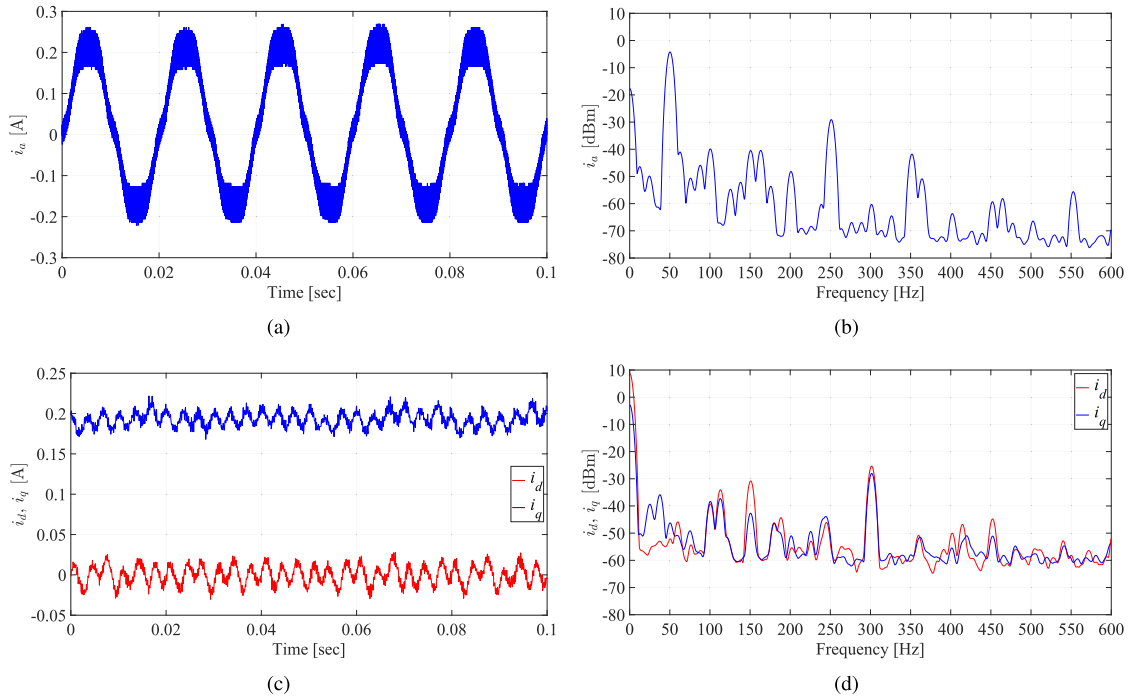


FIGURE 5. Experimental results of the second case with PI controller, feedforward compensator, and NDC only for dead-time disturbance voltage: (a) a-phase current, (b) magnitude spectrum of a-phase current, (c) dq -axes currents, and (d) magnitude spectra of dq -axes currents.

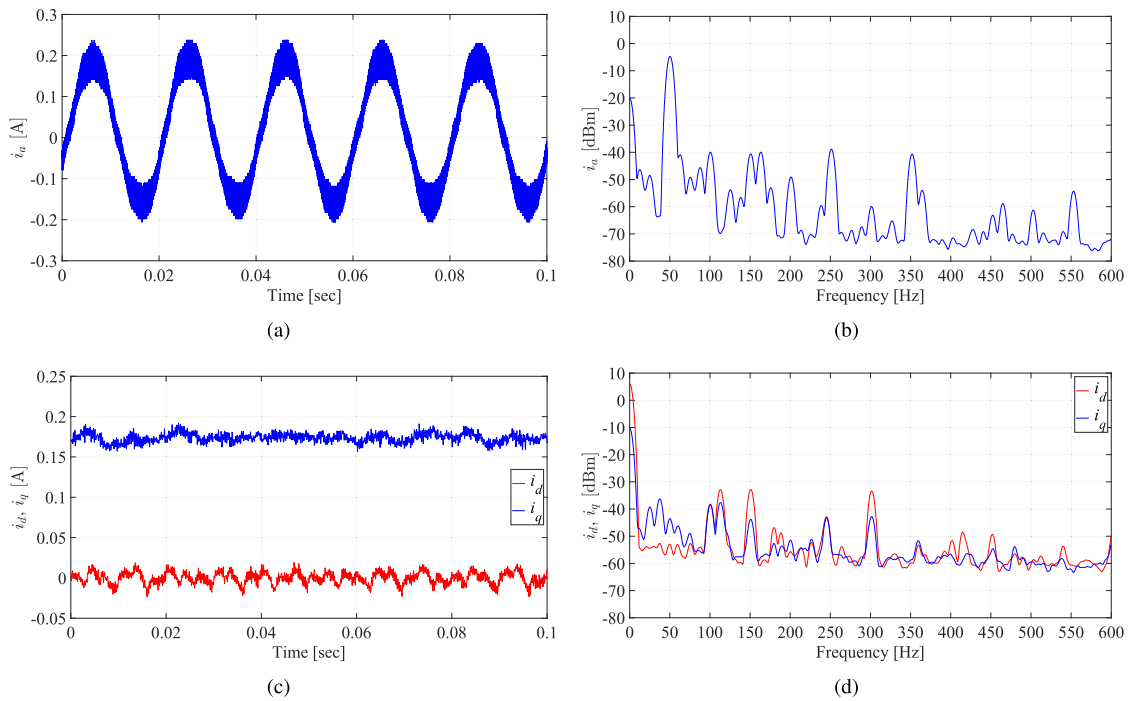


FIGURE 6. Experimental results of the third case PI controller, feedforward compensator, and complete NDC for both dead-time disturbance voltage and parameter errors: (a) a-phase current, (b) magnitude spectrum of a-phase current, (c) dq -axes currents, and (d) magnitude spectra of dq -axes currents.

voltage and the parameter uncertainty. Furthermore, comparing Figs. 5 and 6, it is verified that not only the dead-time disturbance voltage but also the parameter uncertainty must

be compensated for to achieve a high degree of control performance for the dq -axes currents. To more clearly highlight the performance difference among the three experimental

TABLE 3. Total harmonic distortion and steady-state error.

		Case 1	Case 2	Case 3
THD (%)	i_a, i_b, i_c	4.67	3.85	2.8
	i_d	3.23	2.69	1.67
	i_q	3.02	2.49	1.37
SSE in RMS (A)	i_d	0.032	0.028	0.017
	i_q	0.023	0.014	0.011

cases, the THDs of the three-phase and dq -axes stator currents and the SSEs of the dq -axes stator currents are summarized in Table 3, where the SSE is given by the RMS values of 2000 samples of each current signal in the steady state. In Table 3, the remarkable NDC performance is demonstrated as the three-phase current THDs are improved by 40%, the THD of i_d by 48%, the THD of i_q by 55%, the SSE of i_d by 47%, and the SSE of i_q by 52%.

VI. CONCLUSION

An elaborate dynamic model of SPMSMs was developed by considering the VSI dead-time disturbance voltage and parameter uncertainty, which revealed that both the dead-time disturbance voltage and the parameter uncertainty must be compensated to achieve precise dq -axes stator current control. The NDC was designed to compensate for both the dead-time disturbance voltage and parameter uncertainty of the SPMSM drive systems based on the conventional FOC system with the MTPA strategy. The significant advantage of the designed NDC is that it is constructed in a unified form to compensate for both the dead-time disturbance voltage and parameter uncertainty and is implemented with only the static feedback of available system variables without requiring any high-order filters, estimators, or observers. After applying the NDC with the PI controller and feedforward compensator, the closed-loop system is proven to be globally exponentially ultimately bounded with arbitrarily small bounds and fast decay rates that are tuned by adjusting the PI controller and NDC gains. The experimental results show that the designed NDC significantly reduces the THD and SSE of three-phase and dq -axes stator currents, which verifies the usefulness and effectiveness of the designed NDC.

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